

CONDITIONAL PDFS (CONTINUOUS)

LAST TIME:

- CONDITIONAL
PMF

$$P(Y = y_k \mid X = x_j) = P_{Y|X}(y_k | x_j)$$

- CLASS-CONDITIONAL
PDF

$$f_{Y|X}(y \mid x_k)$$

↑ CONTINUOUS ↑ DISCRETE (LABEL)

- ~~Joint~~
CONDITIONAL
PDF

$$f_{Y|X}(y \mid x)$$

↑ BOTH CONTINUOUS

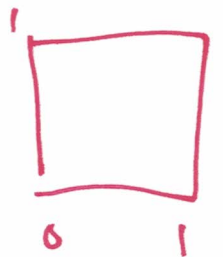
THE CONDITIONAL IS A PDF:

$f_{Y|X}(y | X=x)$ IS A

VALID PDF IN Y , INTEGRATES TO 1,
ETC.

EX $f_{XY}(x,y) = \begin{cases} \frac{3}{4}(2x+y^2) & x,y \in [0,1] \\ 0 & \text{ELSE.} \end{cases}$

COMPUTE $f_{Y|X}(y|x)$.



FIRST STEP: MARGINAL IN X :

$$\begin{aligned} f_X(x) &= \int_0^1 \frac{3}{4}(2x+y^2) dy \\ &= \left[\frac{3}{2}xy + \frac{1}{4}y^3 \right]_{y=0}^{y=1} = \frac{3}{2}x + \frac{1}{4} \end{aligned}$$

$x \in [0, 1]$
0 OTHERWISE.

$$\begin{aligned} \text{So } f_{Y|X}(y|x) &= \frac{\frac{3}{4}(2x+y^2)}{\frac{3}{2}x + \frac{1}{4}} && \frac{\text{JOINT}}{\text{MARGINAL.}} \\ &= \frac{6x + 3y^2}{6x + 1} && x, y \in [0, 1] \end{aligned}$$

FOR EXAMPLE, SAY WE OBSERVED $X = \frac{1}{2}$.

$$f_{Y|X}(y | X = \frac{1}{2}) = \frac{3 + 3y^2}{4} = \frac{3}{4}(1 + y^2)$$

SANITY CHECK:

$$\int_0^1 \frac{3}{4}(1 + y^2) dy = \left[\frac{3}{4}y + \frac{1}{4}y^3 \right]_{y=0}^{y=1} = 1$$

IN GENERAL,

$$\int_0^1 \frac{6x + 3y^2}{6x + 1} dy = \frac{1}{6x + 1} \cdot \left[\underset{\substack{\downarrow \\ 6x + 1}}{6xy + y^3} \right]_{y=0}^{y=1} = 1.$$

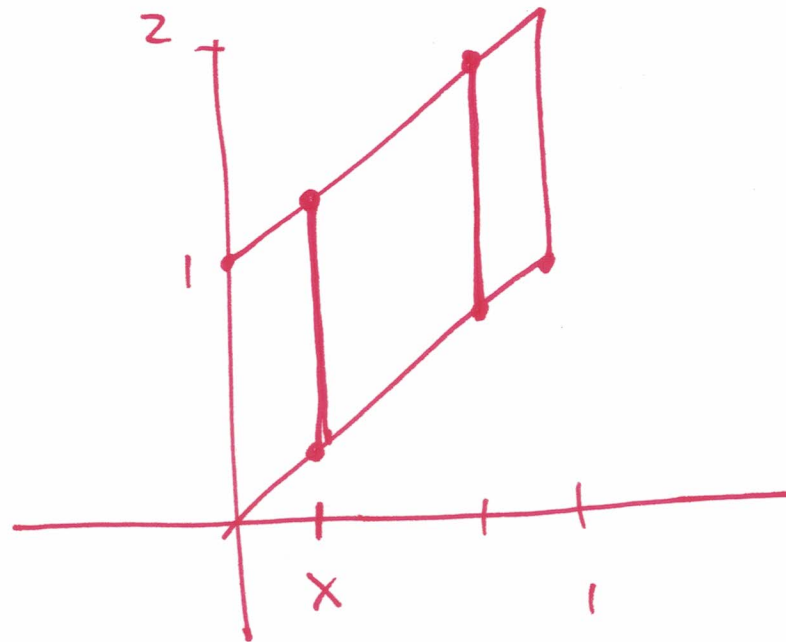
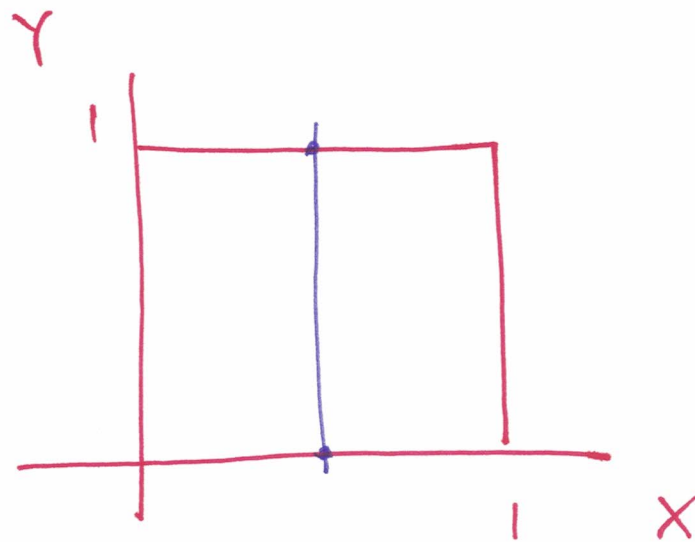
TYPICAL PROBLEM:

GIVEN $f_{Y|X}$, f_X , COMPUTE f_{XY}
 f_Y .

a) $f_{XY} = f_{Y|X} f_X$

b) $\int_{-\infty}^{\infty} f_{XY} dx = f_Y(y).$

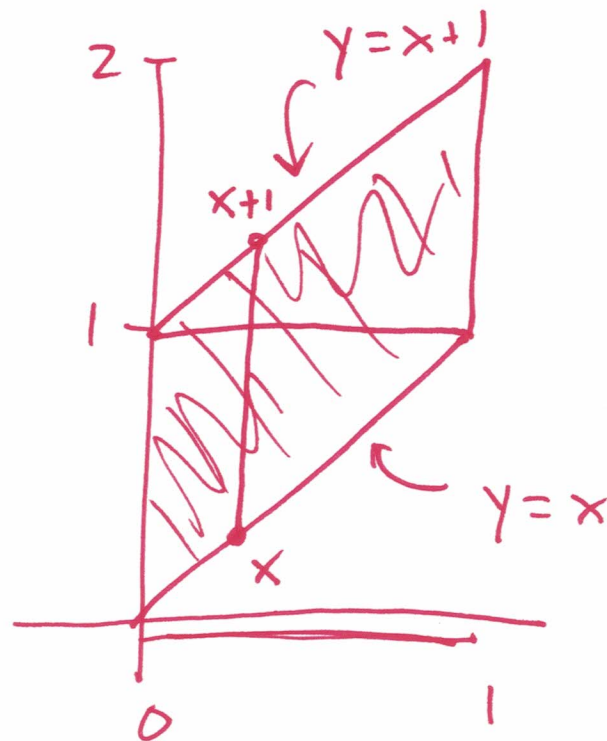
IN EX



$$x \in [0, 1]$$

$$y \in [0, 2]$$

X, Y UNIFORMLY DISTRIBUTED
ON THIS SHAPE:

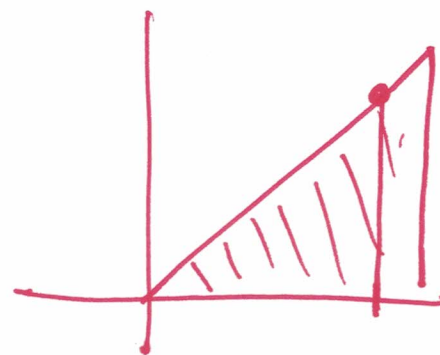


WHAT IS

$$f_{Y|X}?$$

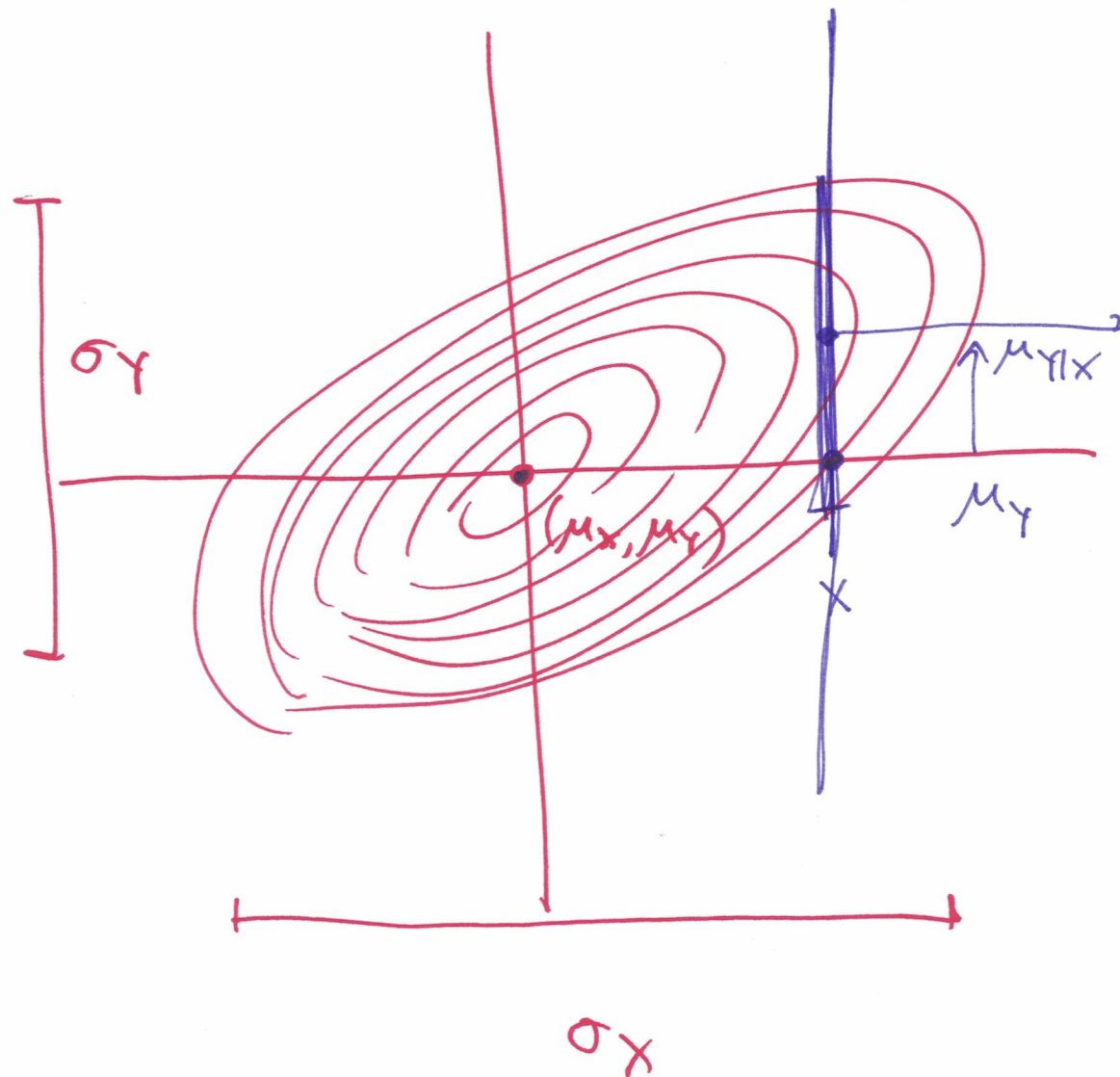
$$= \begin{cases} 1 & y \in [x, x+1] \\ 0 & \text{OTHERWISE} \end{cases}$$

$$\rightarrow \frac{1 \quad y \in [x, x+1]}{1 \quad x \in [0, 1]}$$



IN PARTICULAR FOR THE JOINT GAUSSIAN:

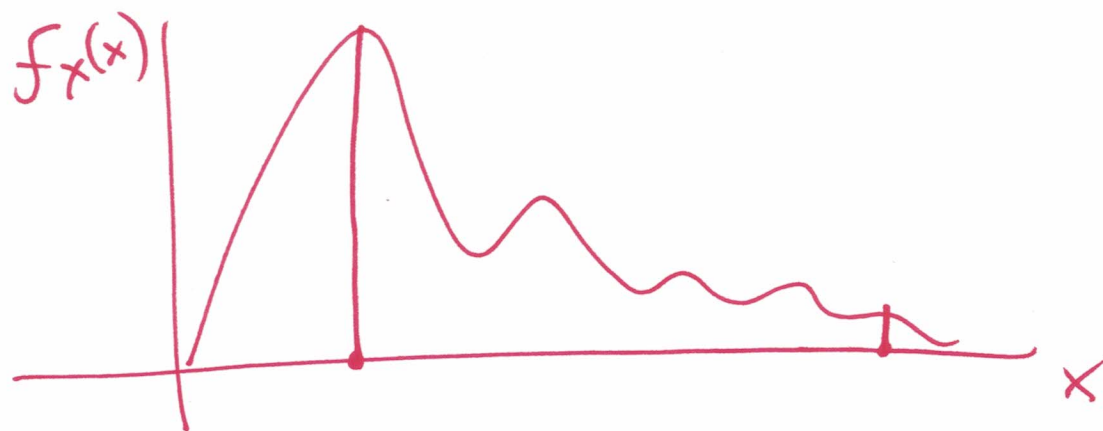
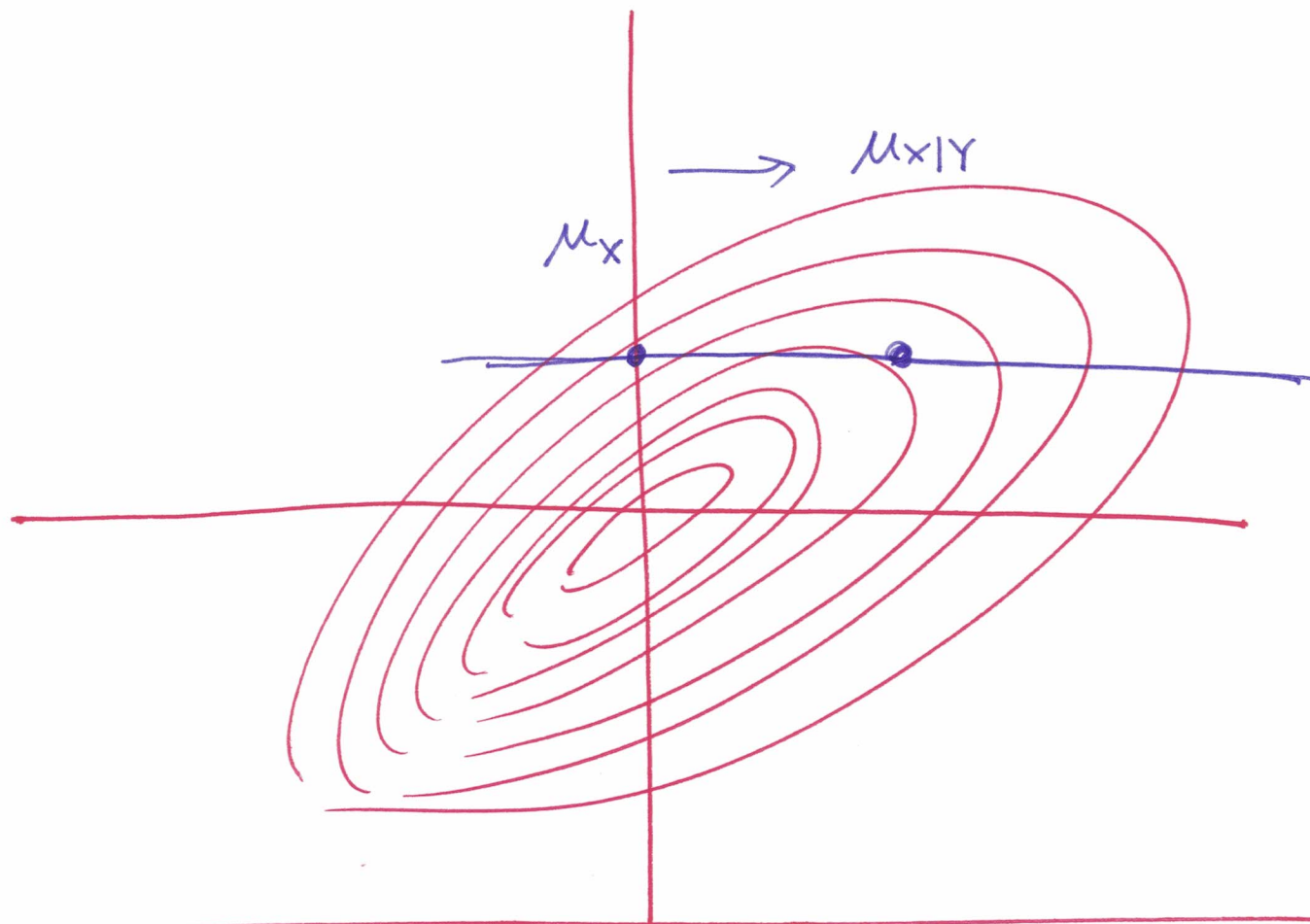
$\mu_x, \mu_y, \sigma_x, \sigma_y, \rho$



$f_{y|x}$ IS
ALSO GAUSSIAN
WITH

$$\mu_{y|x} = \mu_y + \frac{\sigma_y}{\sigma_x} \rho (x - \mu_x)$$

$$\sigma_{y|x} = \sqrt{1 - \rho^2} \sigma_y$$



REVISITING ESTIMATION PROBLEMS:

PARAMETER ESTIMATION.

- 2 WAYS:
- MAXIMUM LIKELIHOOD ESTIMATION (ML)
NO PRIOR
 - MAXIMUM A POSTERIORI (MAP)
(BAYESIAN) USE A PRIOR.

TYPICAL EXAMPLE: FITTING A DISTRIBUTION
TO DATA.

EX: X IS EXPONENTIAL P.V. BUT
WE DON'T KNOW λ ! ALL WE
HAVE ARE SAMPLES FROM THE PDF.

WE SEE SAMPLES $\{x_1, x_2, \dots, x_N\}$
INDEPENDENTLY DRAWN FROM THE PDF.

WHAT IS THE ML ESTIMATE OF
THE PARAMETER λ ?

$$f_{x_i}(x_i | \lambda) = \lambda e^{-\lambda x_i}$$

$\uparrow$ NOT KNOWN. $\uparrow$

N INDEPENDENT SAMPLES:

JOINT PDF OF ALL
SAMPLES

INDIVIDUAL
PDFS SINCE
INDEPENDENT.

$$f(x_1, x_2, \dots, x_N | \lambda) = (\lambda e^{-\lambda x_1}) \dots (\lambda e^{-\lambda x_N})$$

$$= \lambda^N e^{-\lambda(x_1 + x_2 + \dots + x_N)}$$

$$= \lambda^N e^{-\lambda \sum_{i=1}^N x_i}$$

THE ML ESTIMATE: MAKE THIS AS BIG
AS POSSIBLE (AS A FUNCTION OF λ)

$$\frac{d}{d\lambda} f(\lambda) = 0 \quad \text{TO MAXIMIZE.}$$

$$N \lambda^{N-1} e^{-\lambda \sum_{i=1}^N x_i} + (\lambda^N) \left(-\sum_{i=1}^N x_i\right) e^{-\lambda \Sigma} = 0$$

$$N \lambda^{N-1} - \lambda^N \sum_{i=1}^N x_i = 0$$

$$N = \sum_{i=1}^N x_i$$

$$\lambda = \frac{N}{\sum_{i=1}^N x_i} = \frac{1}{\text{MEAN OF } \{x_i\}} = \frac{1}{\boxed{\frac{1}{N} \sum_{i=1}^N x_i}}$$

SAMPLE MEAN.

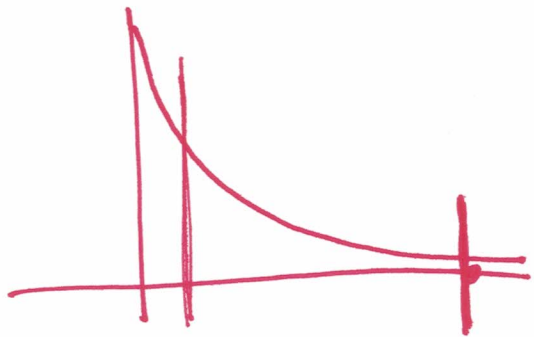
MAKES SENSE! MEAN OF EXPONENTIAL DISTRIBUTION IS $\frac{1}{\lambda}$.

NOW SUPPOSE WE ALSO HAVE A

PRIOR DISTRIBUTION ON THE VALUE OF λ :

$$f_{\lambda}(\lambda) = 2e^{-2\lambda} \quad \lambda > 0$$

↑ EXPONENTIAL PDF WITH MEAN $\frac{1}{2}$.



WHAT IS THE BAYESIAN / MAP
ESTIMATE OF λ NOW?

$$\text{ML:} \quad \max_{\lambda} f(x | \lambda)$$

$$\text{MAP:} \quad \max_{\lambda} f(\lambda | x)$$

$$= \frac{\boxed{f(x | \lambda)} \boxed{f(\lambda)}}{\boxed{f(x)}}$$

LIKELIHOOD PRIOR
 (points to $f(x | \lambda)$) (points to $f(\lambda)$)

DOESN'T DEPEND ON λ
 (points to $f(x)$)

SO WE WANT TO

$$\max_{\lambda} \lambda^N e^{-\lambda \sum_{i=1}^N x_i} \cdot 2e^{-2\lambda}$$

$f(x | \lambda)$ • $f(\lambda)$

$$= 2\lambda^N e^{-\lambda \left(\sum_{i=1}^N x_i + 2 \right)}$$

$$\frac{d}{d\lambda} f(\lambda|x) = 0$$

$$-2\lambda^N (\Sigma + 2) e^{-\lambda(\Sigma + 2)}$$

$$\Rightarrow 2N\lambda^{N-1} e^{-\lambda(\Sigma + 2)}$$

$$\cancel{2\lambda^N e^{-\lambda(\Sigma + 2)}} = 0$$

$$N \cancel{e^{-\lambda(\Sigma + 2)}}$$

$$-\lambda(\Sigma + 2) = 0$$

$$\lambda = \frac{N}{\sum_{i=1}^N x_i + 2}$$

$$= \frac{1}{\frac{1}{N} \sum_{i=1}^N x_i + \frac{2}{N}}$$

PRIOR BELIEF NUDGES
THE ESTIMATE AWAY FROM ML

COMBINING 2 ESTIMATES

IN THIS CASE (AND USUALLY)

$$F(-|x) = \frac{F(x|-) F(-)}{F(x)}$$

⚡ DOESN'T
DEPEND ON
PARAMETER
SO WE
DON'T NEED
TO ACTUALLY
COMPUTE IT.